MIL-L-13751C
23 February 1984
SUPERSEDING
MIL-L-13751B(EA)
25 January 1977

MILITARY SPECIFICATION

LACTOSE, TECHNICAL (METRIC)

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one technical grade of lactose referred to commercially as "crude lactose."

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1168 - Ammunition Lot Numbering

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, Armament Research and Development Center, US Army Armament, Munitions and Chemical Command, ATTN: DRSMC-TSC-S(A), Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6810

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(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

UNIFORM FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

ASTM STANDARDS

D1193 - Reagent Water
D3951 - Commercial Packaging
E11 - Wire-Cloth Sieves For Testing Purposes

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS (AOAC)

Official Methods of Analysis, Twelfth Edition, 1975

(Application for copies should be addressed to the Association of Official Analytical Chemists, P.O. Box 540, Benjamin Franklin Station, Washington, DC 20044.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Chemical and physical characteristics. Lactose shall conform to the chemical and physical characteristics of table I when tested as specified therein.

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be furnished with each transmitter.

TABLE I. Chemical and physical characteristics

Characteristic	Minimum	Maximum	Test paragraph
Lactose, percent by weight	98	--	4.2.4.1
Moisture, percent by weight	--	0.50	4.2.4.1
Ash, percent by weight	--	0.50	4.2.4.1
Nitrogen, percent by weight	--	0.16	4.2.4.1
pH of 5-percent solution	6.0	8.5	4.2.4.1
Specific rotation of 5-percent solution, degrees	51.8	52.8	4.2.4.2
Percent by weight passing through a 2.00 mm sieve	100	--	4.2.4.3

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the lactose produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3). Each lot shall be identified and controlled in accordance with MIL-STD-1168.

4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For test. Sampling shall be conducted in accordance with table II. A representative specimen of approximately 250 (g) shall be removed from each sample container and placed in a clean, dry container labeled to identify the lot and container from which it was taken.

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4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one filled unit container, ready for shipment. Sample unit containers shall be examined for the following defects using an AQL of 1.0 percent defective:

(a) Contents per container not as specified
(b) Container damaged or leaking
(c) Unitization not as specified
(d) Marking incorrect, missing, or illegible

4.2.3.2 For test. Each sample specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.2.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 AOAC test methods. Determine lactose, moisture, ash, nitrogen, and pH of 5-percent solution in accordance with the applicable AOAC test paragraphs listed in table III.

TABLE III. AOAC test methods

Characteristic	AOAC test paragraph no.	Remarks
Lactose	31.063	---
Moisture	31.005	Use a 60°C maximum drying temperature
Ash	31.012	Omit the addition of olive oil
Nitrogen	2.049	---
pH of 5 percent solution	10.030	Use a 5-percent weight per volume solution instead of undiluted samples

4.2.4.2 Specific rotation of 5-percent solution. Weigh 10.000 ± 0.001 g of specimen into a 200-milliliter (mL) volumetric flask and dissolve in water. Add 0.2 mL of ammonium hydroxide solution which contains 10-percent ammonia. Dilute to the volume mark with water. Filter the solution and discard the first 25 mL of filtrate. Polarize the 5-percent solution at 20°C.

4.2.4.3 Granulation. Nest a 2.00-millimeter (mm) sieve conforming to ASTM E11 on a receiving pan. Weigh to the nearest 0.01 g approximately 100 g of the specimen, place on the sieve, cover, and shake for 5 minutes by means of a mechanical shaker geared to produce 300 ± 15 gyrations and 150 ± 10 taps of the striker per minute. Weigh the quantity retained on the sieve and calculate the percent by weight passing through the 2.00-mm sieve.

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5. PACKAGING

5.1 Unit packing, industrial. A quantity of 50 (+0.5 or -0) kilograms of lactose shall be unit packed industrially in accordance with ASTM D3951.

5.2 Packing, industrial. Lactose, unit packed as specified in 5.1, shall require no further protection for shipment other than unitization.

5.2.1 Unitization. Uniform quantities of packs of lactose shall be unitized in a manner to assure protection of the lactose and acceptance by common carrier, and in accordance with Uniform Freight Classification Rules, National Motor Freight Classification Rules and the rules of any other intended mode of transportation. Unitized loads shall be designed to assure safe and stable stacking.

5.3 Marking. Unit packs shall be marked in accordance with ASTM D3951. Unitized loads shall be marked to assure ease of identification and prompt delivery to the intended destination. Each unit pack shall be marked to show lot or batch number and date of manufacture of the lactose.

6. NOTES

6.1 Intended use. Lactose is intended for use in burning-type smoke mixtures.

6.2 Ordering data. Acquisition documents should specify the title, number, and date of this specification.

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E29.

Custodian:

Army - EA

Review activities:

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Preparing activity:

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-L-13751C	LACTOSE, TECHNICAL (METRIC)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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